

Skin Involvement in Disseminated Infection

The skin is involved as a focus in one-third of cases with disseminated infection. More rarely, the skin is the primary location of infection.

Several types of skin lesions have been described, including: - Abscesses with sinus tract formation - Subcutaneous nodules with pustules - Mycetoma - Sporotrichoid nodules - Cellulitis

Diagnosis is based on the demonstration of Gram-positive, partially acid-fast, branching bacilli in tissue or tissue exudate, or by tissue culture—although organism growth often takes several weeks. The treatment of choice is **trimethoprim-sulfamethoxazole**. Other effective antibiotics include **minocycline**, **carbapenems**, and **amikacin**. Transplant patients may require long-term prophylactic therapy to prevent recurrence. Incision and drainage of cutaneous abscesses should also be performed.

Bacillary Angiomatosis

Bacillary angiomatosis is caused by *Bartonella henselae* or *B. quintana* and typically occurs in AIDS patients and other immunocompromised individuals. Cutaneous lesions appear as painful, dome-shaped vascular papules and nodules, often resembling pyogenic granulomas. Disseminated infection can involve the liver, spleen, bone marrow, and brain. A history of cat scratches or bites is common, as cats are the natural reservoir for these bacteria.

Diagnosis is confirmed by identifying pleomorphic bacilli in tissue specimens using Warthin-Starry silver stain. Preferred treatments include oral **erythromycin** or **doxycycline**.

Other Bacterial Infections

Cellulitis caused by *Streptococcus pyogenes*, *Streptococcus pneumoniae*, or *Staphylococcus aureus* may progress rapidly and lead to **necrotizing fasciitis** in immunosuppressed patients. Solid organ transplant recipients may develop

recurrent cellulitis at the elbow—termed "transplant elbow"—often attributed to staphylococcal infection.

Individuals with complement deficiencies (particularly of late-phase components C5–C9) or alcoholism are susceptible to *Neisseria meningitidis* infection. These patients present with acute septicemia, meningitis, disseminated intravascular coagulation, and widespread petechiae and purpura.

Patients with underlying liver disease (e.g., alcoholic cirrhosis or hemochromatosis) are at risk for *Vibrio vulnificus* infection, a Gram-negative bacillus found in seawater and seafood. Infection occurs via ingestion of contaminated shellfish or through skin inoculation from seawater exposure. Clinical presentation includes rapidly progressive septicemia and painful cellulitis, bullae, or ulcers—typically on the lower extremities.

Capnocytophaga canimorsus, a commensal organism in dog and cat saliva, is transmitted through bites or scratches. High-risk hosts include alcoholics, asplenic individuals, and those on glucocorticosteroids. Skin manifestations include macules, papules, purpura, and gangrene. In immunosuppressed patients, *Salmonella enteritidis* has been associated with cutaneous abscesses of the head and neck.

Fungal Infections

Candidiasis

Although mucocutaneous candidiasis is less severe than disseminated disease in the context of acute immunosuppression, it is the most common complication in individuals with chronic cell-mediated immune dysfunction. Approximately 10% of patients with such immune impairment develop clinically significant mucocutaneous candidiasis.